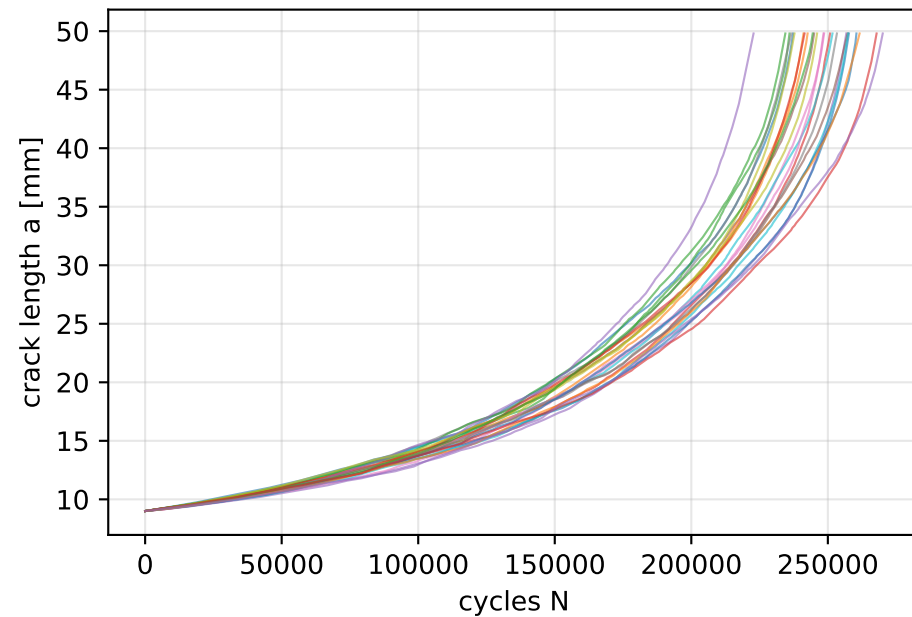
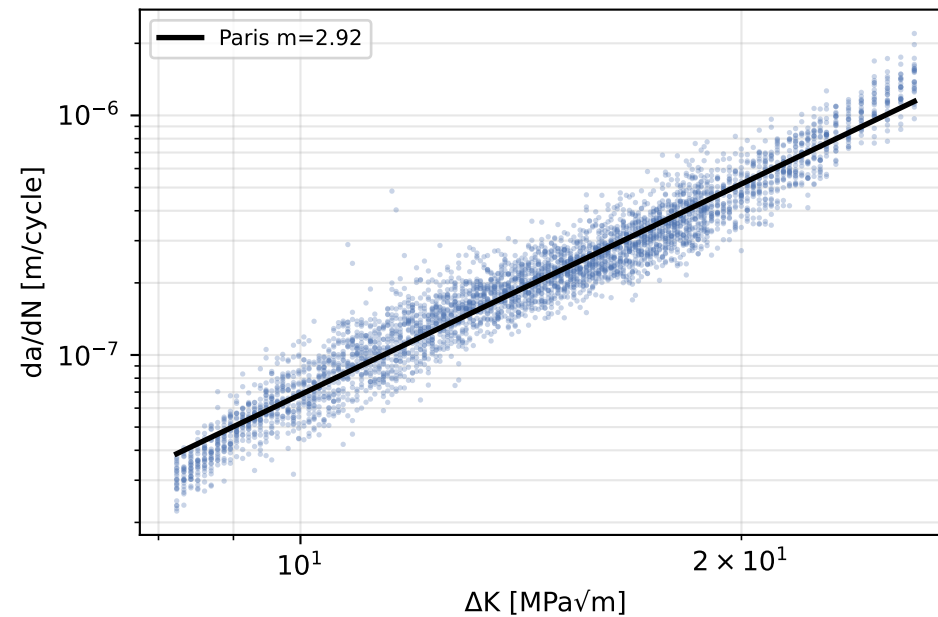


Physics life-clock validated on REAL crack-growth data (Virkler 2024-T3, 25 specimens) — identifiable RUL where the data-driven HI is not

(1) REAL crack growth — Virkler 2024-T3
25 specimens, $a(N)$



(2) da/dN - ΔK on REAL data: clean Paris law
($m \approx 2.9$ = textbook 2024-T3)



(3) On REAL crack data the physics clock is
IDENTIFIABLE (vs the data HI)

